

MATH 112B

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1.1 Lecture 1 - Gradient Systems

Notation: for a function with two variables

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$
$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Linearity.

$$\frac{\partial}{\partial x} cf = c \frac{\partial}{\partial x} f \tag{1}$$

$$\frac{\partial}{\partial x} (f + g) = \frac{\partial}{\partial x} f + \frac{\partial}{\partial x} g \tag{2}$$

Gradient. For $f = f(x, y)$, define the gradient as the following:

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

Properties:

1. Magnitude: $|\vec{\nabla} f| = \sqrt{f_x^2 + f_y^2}$
2. Direction: $\vec{u} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|} = \text{unit vector in the } (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}) \text{ direction.}$
3. $\vec{\nabla} f$ at point (x_0, y_0) is $\perp$ to the level set at point (x_0, y_0) .
4. $\vec{\nabla} f$ is defined at all the points (x, y) where partials exists.
5. A collection of all $\vec{\nabla} f$ is a vector field called gradient field.
6. One could verify $F(M(x, y), N(x, y))$ is a gradient field by verifying

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

where $F = \vec{\nabla} f$

1.2 Lecture 2 - Conservation Laws

Conservation Laws

This is used to describe: bacterial movement in a petridish, nerve signal traveling through an axon, heat diffusion in a rod, cars on the highway.

We start with the following assumptions:

- Motion happens in 1D.
- Cross-sectional area is constant.

Consider a 1D tube, with particles inside. How do we describe the change of concentration of particles?

$$\begin{array}{c} \text{change in pop between} \\ x \text{ and } x + \Delta x \end{array} = \begin{array}{c} \text{rate of entry into} \\ (x, x + \Delta x) \\ \text{per unit time} \end{array} - \begin{array}{c} \text{rate of exit into} \\ (x, x + \Delta x) \\ \text{per unit time} \end{array}$$

Define

- $c(x, t) = \text{concentration of particles at } x \text{ at time } t$

- $J(x, t)$ = flux of particles at (x, t) = # of particles crossing a unit area at x in the positive direction, per unit time.
- A = cross-sectional area of the tube.
- ΔV = volume of our segment = $A \cdot \Delta x$

We have

$$\begin{aligned} \frac{\partial}{\partial t} \underbrace{(c(x, t) \cdot A \cdot \Delta x)}_{\text{\# of particles}} &= J(x, t)A - J(x + \Delta x, t)A \\ \frac{\partial c}{\partial t} &= \frac{J(x, t) - J(x + \Delta x, t)}{\Delta x} \\ \lim_{\Delta x \rightarrow 0} \frac{\partial c}{\partial t} &= \lim_{\Delta x \rightarrow 0} \frac{J(x, t) - J(x + \Delta x, t)}{\Delta x} \\ \frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial x} \leftarrow \text{Conservation of mass} \end{aligned}$$

Consider a new term,

$$\begin{array}{ccccc} \Delta \text{pop between} & & \text{rate of entry into} & \text{rate of exit into} & \text{rate of local} \\ x \text{ and } x + \Delta x & = & (x, x + \Delta x) & - & (x, x + \Delta x) \pm & \text{creation} \\ & & \text{per unit time} & & \text{per unit time} & \text{or degradation} \end{array}$$

We now have

- $\sigma(x, t)$ = # of particles created/removed per unit volume at time t

Now our conservation laws become

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} \pm \sigma(x, t)$$

To generalize to 2D or 3D:

$$\frac{\partial c}{\partial t} = -\vec{\nabla} \cdot \vec{J} \pm \sigma(\vec{x}, t)$$

Now we have $c = c(\vec{x}, t)$

$$\sigma = \sigma(\vec{x}, t)$$

$$\vec{J} = (J_x, J_y, J_z)$$

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

So we have

$$\frac{\partial c}{\partial t} = -\left(\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}\right) \pm \sigma(x, y, z, t)$$

[ex] Consider a moving fluid + organisms that move with the flow (not actively). Suppose that the velocity of the fluid is given by

$$\vec{v}(x, y, z, t)$$

Let $c(x, y, z, t)$ = concentration of organism.

What is the flux?

$$\begin{aligned} \vec{J} &= c \cdot \vec{V} \quad \text{Assume } \sigma = 0 \\ \Rightarrow \frac{\partial c}{\partial t} &= -\left(\frac{\partial c V_x}{\partial x} + \frac{\partial c V_y}{\partial y} + \frac{\partial c V_z}{\partial z}\right) \end{aligned}$$

In 1D:

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}(cV) \\ &= -\frac{\partial c}{\partial x}V - \frac{\partial V}{\partial x}c \end{aligned}$$

What kind of PDE is this?

- 1st order
- 2 variables: x, t
- Linear
- Transport equation

[ex] Attraction/repulsion. Suppose that ψ is a source of attraction for the bacteria. The motion is determined by the gradient of ψ .

In 1D: $\psi = \psi(x, t)$, $J = \frac{\partial \psi}{\partial x}$

$$J = c\alpha \vec{\nabla} \psi$$

We have

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial X} \\ \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial t} \left(c\alpha \frac{\partial \psi}{\psi x} \right)\end{aligned}$$

ψ is given.

$$\frac{\partial c}{\partial t} = -\alpha \frac{\partial c}{\partial x} \cdot \frac{\partial \psi}{\partial x} - \alpha c \frac{\partial c}{\partial x} \cdot \frac{\partial^2 \psi}{\partial x^2}$$

This is 1st order, linear PDE in 2 variables.

ex Temperature of a heated rod.

Here we define $c(x, t)$ as concentration of energy/temperature. And since heat spreads from hot to cold, we get the following.

Fick's Law : $J = -D \cdot \frac{\partial c}{\partial x}$

Then we have

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x} \left(-D \frac{\partial c}{\partial x} \right) \\ &= D \frac{\partial^2 c}{\partial x^2} \leftarrow \text{Heat equation}\end{aligned}$$

In higher dimensions:

$$\begin{aligned}\frac{\partial c}{\partial t} &= \vec{\nabla} (D \vec{\nabla} c) \\ &= D \left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2} \right)\end{aligned}$$

ex Random motion of particles.

Denote λ_r as the amount of particles jumping from $x \rightarrow x + \Delta x$

Denote λ_l as the amount of particles jumping from $x \rightarrow x - \Delta x$

Denote $c(x, t) = \#$ of particles within $(x, x + \Delta x)$ at time t :

$$c(x, t + \Delta t) = c(x, t) + \lambda_r \cdot c(x - \Delta x, t) + \lambda_l \cdot c(x + \Delta x, t) - \lambda_l c(x, t) - \lambda_r c(x, t)$$

Using Taylor Series:

$$\begin{aligned}
 c(x, t + \Delta t) &= c(x, t) + \frac{\partial c}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 c}{\partial t^2} (\Delta t)^2 + \dots \\
 c(x + \Delta x, t) &= c(x, t) + \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 + \dots \\
 c(x - \Delta x, t) &= c(x, t) - \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 + \dots
 \end{aligned}$$

We can simplify using $\lambda_r = \lambda_l = \lambda$, then we have

$$\begin{aligned}
 c + \frac{\partial c}{\partial t} \Delta t + \frac{\partial^2 c}{\partial t^2} \frac{(\Delta t)^2}{2} + \dots &= c \\
 &+ \lambda \left(c + \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 \right) \\
 &+ \lambda \left(c - \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 \right) \\
 &- \lambda c - \lambda c \\
 \frac{\partial c}{\partial t} \Delta t + \frac{\partial^2 c}{\partial t^2} \frac{(\Delta t)^2}{2} &= \lambda \frac{\partial^2 c}{\partial x^2} (\Delta x)^2
 \end{aligned}$$

Call $\lim_{\Delta x \rightarrow 0, \Delta t \rightarrow 0} \frac{(\Delta x)^2}{\Delta t} \equiv K$ and ignore second order time term as we take the limit.

$$\begin{aligned}
 \frac{\partial c}{\partial t} \Delta t &= \lambda (\Delta x)^2 \cdot \frac{\partial^2 c}{\partial x^2} \\
 \frac{\partial c}{\partial t} &= \lambda K \cdot \frac{\partial^2 c}{\partial x^2} \leftarrow \text{Diffusion equation}
 \end{aligned}$$

1.3 Lecture 3 - Conservation Laws cont.

Recall

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} + \sigma(x, t)$$

ex Propagation of Action Potential along an axon (neural cell). Let

- x = the distance along the axon.

- $q(x, t)$ = the charge density per unit length.
- $J(x, t)$ = flux of charged particles
- $\sigma(x, t)$ = rate of charge entering/leaving

We have the following transport equation for the charge:

$$\frac{\partial q}{\partial t} = -\frac{\partial J}{\partial x} + \sigma(x, t)$$

So what is $J(x, t)$? It is the electrical current.

For $J(x, t)$, we use Ohm's Law:

$$J = -a \frac{\partial V}{\partial x}$$

where a is a constant and $\frac{\partial V}{\partial x}$ is the voltage gradient.

This relates charge to voltage:

$$q = bV$$

where q is charge, b is a constant, V is voltage.

We then have

$$\begin{aligned} \frac{\partial}{\partial t}(bV) &= -\frac{\partial}{\partial x}\left(-a \frac{\partial V}{\partial x}\right) + \sigma(x, t) \cdot \frac{1}{b} \\ \underbrace{\frac{\partial V}{\partial t}}_{\text{diffusion eq}} &= \frac{a}{b} \frac{\partial^2 V}{\partial x^2} + \frac{\sigma(x, t)}{b} \end{aligned}$$

Generalize this to 2d or 3d

$$\frac{\partial c}{\partial t} = D \left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2} \right) \quad (\text{diffusion eq})$$

Del operator: (note that $\vec{\nabla} u = \text{grad } u$)

$$\begin{aligned} \vec{\nabla} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \\ u(x, y, z) &\rightarrow \vec{\nabla} \rightarrow \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right) \\ \text{scalar} &\rightarrow \vec{\nabla} \rightarrow \text{vector} \end{aligned}$$

Divergence:

Consider the dot product with $\vec{V} = (V_x, V_y, V_z)$, we have

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = \text{divergence of } V$$

vector $\rightarrow$ div $\rightarrow$ scalar

Laplacian:

Consider the combination of div and grad

$$\vec{\nabla}(\vec{\nabla}u)$$

where $\vec{\nabla}u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$. We have

$$\underbrace{\vec{\nabla} \cdot \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)}_{\text{div} \cdot \text{grad } u} = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = \underbrace{\Delta u}_{\text{Laplacian } u}$$

$$\Delta = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$$

Back to the diffusion equation

$$\frac{\partial c}{\partial t} = D\Delta c \quad (\text{diffusion eq})$$

$$\frac{\partial c}{\partial t} = D\Delta c + \sigma \quad (\text{reaction-diffusion eq})$$

where D is the diffusion coefficient.

This begs the question, what are the units of D ?

$$\frac{c}{\text{time}} = D \cdot \frac{c}{\text{space}^2}$$

$$D = \frac{\text{space}^2}{\text{time}}$$

ex How quickly can a cell get oxygen from blood? (for O_2 , $D \approx 0.2\text{cm}^2/\text{sec}$)

1. The approximate distance through which diffusion works in a given time is $\propto \sqrt{Dt}$

2. The approximate time it takes to diffuse a distance d is $\propto \frac{d^2}{D}$

Time it takes O ₂ to diffuse	
Distance	Diffusion time
1 μm	10^{-3} sec
10 μm	0.1 sec
1 mm	10^3 sec $\approx$ 15 mins
1 cm	10^5 sec $\approx$ 25 hr
1 m	10^8 sec $\approx$ 27 yrs

Role of diffusion in cancer:

Cells in the core of the tumor die due to lack of oxygen, forming a necrotic core. There was a paper by Judah Folkman in 1971, that a tumor can only grow up to 2mm in diameter (physics). This leads to tumors developing methods to deliver oxygen to its core like angiogenesis.

Consider the diffusion equation in 1d:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < l, \quad t > 0$$

We wish to find $u(x, t)$. Set the initial condition to $u(x, 0) = f(x)$. Set the boundary condition to $u(0, t) = u(l, t) = 0$, $t > 0$, this is also known as Dirichlet boundary condition.

Alternatively, we could have the Neumann boundary condition, which states that $\frac{\partial u}{\partial x}u(0, t) = \frac{\partial u}{\partial x}u(l, t) = 0$. This means no flux boundary (insulating).

[ex] Consider $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $t > 0$. With initial condition $u(x, 0) = f(x) = 2x^2$. With boundary condition $\begin{cases} u(0, t) = 0 \\ u(1, t) = 2 \end{cases}$. Find the long term behavior $\lim_{t \rightarrow \infty} u(x, t)$.

Assume that as $t \rightarrow \infty$, the solution comes to a fixed profile.

$$\begin{aligned} \frac{\partial u}{\partial t} &= D \frac{\partial^2 u}{\partial x^2} = 0 \\ \text{As } t &\rightarrow \infty, \quad \frac{\partial u}{\partial t} = 0 \end{aligned}$$

We have a boundary value problem:
$$\begin{cases} D \frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow u(x) = cx + b \\ u(0) = 0 \\ u(1) = 2 \end{cases}$$

$$\frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow \frac{dV}{dx} = 0 \Rightarrow V(x) = c$$

Use boundary condition:

$$\begin{aligned} u(0) &= c \cdot 0 + b = 0 \Rightarrow b = 0 \\ u(1) &= c \cdot 1 + 0 = 2 \Rightarrow c = 2 \\ \Rightarrow u(x) &= 2x \end{aligned}$$

To solve PDEs, we need additional techniques.

Suppose $f(x)$ defined on $[0, \pi]$. This function can be represented as a sum of sines and cosines.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where $b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$ and

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$.

This is known as sine and cosine series representation.

1.4 Lecture 4 -